UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 002 · Geometrical interpretation of partial derivatives · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S002·7	Deck 002, slide 7 (rendered .pptx)
T002 00:06:10	AV-summary transcript 002, segment start time
V002 00:07:45	Frame grabbed from the class 002 video
CW U1 P2 · ESA Dec-23 1a	Unit 1 class-work problem 2 · past-paper question

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A Formula sheet

1 Higher-order partial derivatives

E1.1 · First-order partial derivatives ESA ★★

$$f_x = \frac{\partial f}{\partial x}, \qquad f_y = \frac{\partial f}{\partial y}$$

GIVES	Rate of change of f in the x (or y) direction.
APPLIES	Differentiate once; hold the other variable constant.
NOTE	Two of them for a function of two variables.

S002·3 · T002 00:00:44

E1.2 · Second-order partial derivatives ESA ★★

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y\,\partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x\,\partial y}$$

GIVES The four second-order partials of $f(x, y)$.

APPLIES Differentiate twice. In f_{xy}, the subscripts are read left to right: first x, then y.

NOTE Notation order varies between books. The two mixed partials are equal when continuous (E1.4).

ADDED FOR CLARITY S002·3 · T002 00:00:58

E1.3 · How many partial derivatives of order n? ESA ★

2 variables: 2^n m variables: m^n

GIVES The number of nth-order partial derivatives, counting each order of differentiation separately.

APPLIES Two variables: 2 first-order, 4 second-order, 8 third-order.

NOTE With equal mixed partials, two variables give only $n + 1$ distinct ones.

S002·3 · T002 00:01:37 · T002 00:02:13

E1.4 · Equality of mixed partials (Clairaut) ESA ★★

$$f_{xy} = f_{yx} \quad \Longleftrightarrow \quad \frac{\partial^2 f}{\partial y\,\partial x} = \frac{\partial^2 f}{\partial x\,\partial y}$$

GIVES Order of differentiation does not matter.

APPLIES When f_{xy} and f_{yx} are continuous (true for every function in this unit's problems).

NOTE Verified in CW U1 P2 and P8.

TRANSCRIPT ONLY ADDED FOR CLARITY T002 00:00:24 · T002 00:14:59 · CW U1 P2

2 The one-variable picture: derivative as slope

E2.1 · Derivative as slope (one variable) ESA ★

$$\left.\frac{dy}{dx}\right|_{x_0} = f'(x_0) = \text{slope of the tangent to } y = f(x) \text{ at } (x_0, f(x_0))$$

GIVES Slope of the tangent line.

APPLIES f differentiable at x_0.

S002·4 · T002 00:03:13

E2.2 · Tangent line to y = f(x)

ESA ★

$$y - f(x_0) = f'(x_0)\,(x - x_0)$$

GIVES Equation of the tangent at x_0.

APPLIES f differentiable at x_0.

NOTE Point-slope form.

ADDED FOR CLARITY S002·4

4 First-order partials as slopes of tangents

E4.1 · ∂f/∂x as an ordinary derivative

ESA ★★

$$\left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} = \left.\frac{d}{dx}\big(f(x,y_0)\big)\right|_{x=x_0}$$

GIVES Slope of the tangent to the surface at $(x_0, y_0, f(x_0,y_0))$ that is parallel to the x-axis.

APPLIES Fix $y = y_0$ first; $f(x,y_0)$ depends only on x.

NOTE The tangent lies in the plane $y = y_0$.

S002·7 · V002 00:07:45 · T002 00:06:10

E4.2 · ∂f/∂y as an ordinary derivative

ESA ★★

$$\left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} = \left.\frac{d}{dy}\big(f(x_0,y)\big)\right|_{y=y_0}$$

GIVES Slope of the tangent to the surface that is parallel to the y-axis.

APPLIES Fix $x = x_0$ first; $f(x_0,y)$ "just depends on y".

NOTE The tangent lies in the plane $x = x_0$.

S002·7 · V002 00:07:45 · T002 00:07:10

E4.3 · The two tangent lines at P

ESA ★

$$x\text{-direction: } z - z_0 = f_x(x_0,y_0)\,(x - x_0),\quad y = y_0$$
$$y\text{-direction: } z - z_0 = f_y(x_0,y_0)\,(y - y_0),\quad x = x_0$$

GIVES Equations of the tangents whose slopes are f_x and f_y.

APPLIES $z_0 = f(x_0,y_0)$.

NOTE Point-slope form inside each plane section.

ADDED FOR CLARITY S002·7

5 Slope in any other direction: the gradient

E5.1 · Gradient

ESA ★

$$\nabla f = f_x\,\mathbf{i} + f_y\,\mathbf{j}$$

GIVES Vector whose components are the slopes along the x- and y-axes.

APPLIES Taken at the point (x_0, y_0).

NOTE Covered fully in vector calculus.

S002·8 · T002 00:09:23

E5.2 · Slope along any direction (directional derivative)

ESA ★

$$D_{\hat{u}} f = \nabla f \cdot \hat{u} = a\,f_x + b\,f_y, \qquad \hat{u} = (a, b),\ a^2 + b^2 = 1$$

GIVES Slope of the tangent in the direction $\hat{u}$.

APPLIES $\hat{u}$ must be a unit vector.

NOTE $\hat{u} = \mathbf{i}$ gives f_x; $\hat{u} = \mathbf{j}$ gives f_y. The largest slope is $|\nabla f|$.

ADDED FOR CLARITY S002·8

6 Second-order partials: concavity and mixed partials

E6.1 · Concavity from second partials

ESA ★★

$$f_{xx} > 0 \Rightarrow \text{concave up in } x, \qquad f_{yy} > 0 \Rightarrow \text{concave up in } y$$

GIVES Shape of the surface along each axis direction.

APPLIES $f_{xx} < 0$ (or $f_{yy} < 0$) gives concave down in that direction.

NOTE Same test as d^2y/dx^2 for one variable.

S002·9 · T002 00:11:19

E6.2 · Mixed partial as a change of slope

ESA ★★

$$f_{xy} = \frac{\partial}{\partial y}\left(f_x\right)$$

GIVES How the x-slope f_x changes as you move in the y direction.

APPLIES Likewise $f_{yx} = \partial(f_y)/\partial x$: the change of the y-slope along x.

S002·9 · T002 00:11:45

E6.3 · Names of higher derivatives (motion) ESA ★

$$\frac{dy}{dt} = \text{velocity}, \quad \frac{d^2y}{dt^2} = \text{acceleration}, \quad \frac{d^3y}{dt^3} = \text{jerk}$$

GIVES Physical meaning when the independent variable is time.

APPLIES One-variable analogy only.

NOTE Names exist up to about the 6th derivative.

TRANSCRIPT ONLY T002 00:10:23 · T002 00:10:57

B Symbol glossary

Symbol	Name	Meaning	Appears in
f, z	Function / dependent variable	$z = f(x, y)$; its graph is a surface	E1.1–E6.2
x, y	Independent variables	Coordinates in the xy-plane	all
(x_0, y_0)	Base point	Point where the derivatives are evaluated	E4.1–E5.2
P, z_0	Point on the surface	$P = (x_0, y_0, z_0)$ with $z_0 = f(x_0, y_0)$	E4.3
∂	"Partial", read "dou"	Differentiate one variable, hold the others constant	E1.1–E1.4
f_x, f_y	First-order partials	Slopes parallel to the x- and y-axes	E1.1, E4.1–E5.2
f_{xx}, f_{yy}	Pure second-order partials	Concavity along x and along y	E1.2, E6.1
f_{xy}, f_{yx}	Mixed second-order partials	Change of one slope along the other direction	E1.2, E1.4, E6.2
m, n	Counts	m = number of variables, n = order	E1.3
$f'(x_0)$, dy/dx	Ordinary derivative	Slope of a tangent to a plane curve	E2.1, E2.2
∇f	Gradient ("del f")	$f_x\,\mathbf{i} + f_y\,\mathbf{j}$	E5.1, E5.2
$\mathbf{i}$, $\mathbf{j}$	Unit vectors	Along the x- and y-axes	E5.1
$\hat{u} = (a, b)$	Unit direction vector	$a^2 + b^2 = 1$	E5.2
$D_{\hat{u}} f$	Directional derivative	Slope of the surface in direction $\hat{u}$	E5.2
t	Time (in F14)	Independent variable in the motion analogy	E6.3

Remark. No physical units here: every slope is a pure number (rise per unit run) unless x, y, z carry units. Then f_x has units of z per unit of x.

C Notation equivalents

Subscript	Leibniz	Also written	Read as
f_x	$\dfrac{\partial f}{\partial x}$	z_x, $\dfrac{\partial z}{\partial x}$	slope along x
f_{xx}	$\dfrac{\partial^2 f}{\partial x^2}$	z_{xx}	bending along x
f_{xy}	$\dfrac{\partial^2 f}{\partial y\,\partial x}$	$(f_x)_y$	x first, then y
f_{yx}	$\dfrac{\partial^2 f}{\partial x\,\partial y}$	$(f_y)_x$	y first, then x
∇f	$f_x\,\mathbf{i} + f_y\,\mathbf{j}$	grad f	vector of slopes